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PHYSICS

Noether's Theorem and Why Energy Is Conserved, Except When It Isn't

The 1915 crisis in general relativity that produced the deepest structural result in physics — and the reason cosmologists can say, without embarrassment, that the expanding universe conserves no energy at all.

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In the autumn of 1915, David Hilbert — one of the two or three most powerful mathematicians alive — found something wrong with Albert Einstein's brand-new theory of gravity, and it was not a small thing. General relativity appeared to let energy simply vanish, or appear from nowhere, in violation of a conservation law physicists had trusted since the 1840s. Hilbert and his Göttingen colleague Felix Klein worked on the puzzle themselves and got stuck. So they recruited a 33-year-old mathematician to solve it for them, and she solved something much bigger instead. Her name was Emmy Noether, and the university that had just invited her would not let her hold a paid teaching position, because she was a woman [1].

Three years later she handed Hilbert and Klein a pair of theorems that did not so much patch the hole in general relativity as reveal what conservation laws actually are. Physicists since Newton had treated the conservation of energy, momentum, and angular momentum as separate bedrock facts about nature — three independent gifts, each verified by experiment, each apparently unrelated to the others. Noether showed they were all the same fact wearing different clothes: each one is the shadow cast by a symmetry. Energy conservation exists because the laws of physics do not care what day it is. Momentum conservation exists because they do not care where you are. And nearly a century later, that same logic is the reason cosmologists will tell you, correctly, that the expanding universe as a whole does not conserve energy at all [2].

An Assistant in Name Only

Noether earned her doctorate in 1907 at the University of Erlangen, summa cum laude, under the algebraist Paul Gordan, at a time when German universities barely tolerated women attending lectures, let alone examining them [3]. She spent the next several years working essentially without pay or title, doing research good enough that by 1915 Hilbert and Klein wanted her specifically — not a promising student, but the person who understood invariant theory better than almost anyone in Europe, exactly the mathematical toolkit general relativity needed [3], [4]. They invited her to Göttingen. The philosophy faculty said no: a woman could not be habilitated, the formal credential required to lecture and draw a salary. Hilbert is widely quoted pushing back at a faculty senate meeting with some version

of the line, “I do not see that the sex of the candidate is an argument against her admission — after all, we are a university, not a bathhouse” [5]. As with many best-loved scientific one-liners, the wording has been polished by a century of retelling, but the underlying fact is not in dispute: Hilbert lost the argument. For four years Noether lectured at Göttingen under courses formally listed in Hilbert’s own name, “with the assistance of Dr. E. Noether,” and was paid nothing at all [3].

The problem she had been brought in to fix was genuinely strange. In ordinary physics, energy conservation is a bookkeeping statement: total energy in equals total energy out, everywhere, always. General relativity complicated this because it does not treat gravity as a force acting in a fixed background space — matter and energy actively bend spacetime itself, and the geometry evolves as a dynamical actor in its own right. Try to write down a global energy-conservation law the old way and the curvature terms refuse to cooperate; energy seemed able to slosh into or out of the gravitational field with no clean accounting for where it went [2], [4]. Hilbert wanted to know whether this was a mistake in the theory or a mistake in how physicists were thinking about “energy.” It was the second, and Noether worked out why.

The Theorem: Symmetry Is the Reason

Noether’s 1918 paper, “Invariante Variationsprobleme,” contains two theorems, and it is the first that became one of the load-bearing walls of twentieth-century physics [4], [6]. Stating it requires one piece of machinery that was already standard by 1918: the idea that a system’s entire dynamics can be packed into a single function called the Lagrangian, $L(q, \dot{q}, t)$, built from positions and velocities, and that nature always follows the path along which the action

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt$$

is stationary against small wiggles, $\delta S = 0$. That is the principle of least action, and turning the crank correctly reproduces Newton’s laws, Maxwell’s equations, and general relativity’s field equations alike [6], [7].

Now suppose the system can be transformed in some continuous way — every position nudged by a small amount, the clock shifted, the whole apparatus rotated — and the Lagrangian does not change. That is a symmetry. Noether’s first theorem says that every continuous symmetry of the action corresponds to exactly one conserved quantity, and it hands you a recipe for building that quantity out of the symmetry itself [6], [7]. Feed in different symmetries and the machine manufactures the conservation laws you already knew, this time as consequences rather than axioms.

Symmetry of the action	Conserved quantity	Everyday version
Time translation	Energy	A pendulum's period is the same on Tuesday as on Wednesday
Space translation	Linear momentum	A puck coasts at constant velocity on frictionless ice
Rotation	Angular momentum	A skater spins faster as her arms pull in, total spin unchanged
Phase shift of a charged field	Electric charge	No reaction ever changes the total charge in the room

CHECK THIS AGAINST YOUR INTUITION

Energy conservation has been taught for two centuries as a primitive fact about the world. If it is instead a consequence of something else, what exactly is that something else — and what would have to be true of a physical system for the consequence to fail?

The something else is the claim that the rulebook is the same today as tomorrow: the Lagrangian carries no explicit dependence on t . Energy conservation is downstream of that invariance, not parallel to it, which makes the failure condition uncomfortably concrete. Find a corner of reality where the laws themselves shift from one moment to the next, and energy there simply stops being conserved — no paradox, no missing term, just a theorem whose hypothesis was never satisfied. There is such a corner, and it is the entire universe.

A Derivation You Can Run

The theorem is short enough to derive in half a page. Start from the stationarity of the action. Varying $\mathcal{S}$ and requiring $\delta\mathcal{S} = 0$ for arbitrary variations that vanish at the endpoints gives the Euler–Lagrange equation, the equation of motion for any Lagrangian system:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0.$$

Now consider a different kind of variation: not an arbitrary wiggle, but a specific infinitesimal transformation $q \rightarrow q + \epsilon \delta q$ that we suspect is a symmetry. To first order in ϵ , the Lagrangian changes by

$$\delta L = \frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q}.$$

This is where the equation of motion earns its keep. On any physically realized path, Euler–Lagrange lets us replace $\partial L / \partial q$ with $\frac{d}{dt}(\partial L / \partial \dot{q})$, and since $\delta \dot{q} = \frac{d}{dt} \delta q$, the two terms collapse into a single total derivative:

$$\delta L = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q + \frac{\partial L}{\partial \dot{q}} \frac{d}{dt} \delta q = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \delta q \right).$$

If the transformation is a symmetry — if $\delta L = 0$, or more generally if $\delta L = \dot{K}$ for some function K — then the right-hand side vanishes and we are left with a conservation law:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \delta q - K \right) = 0, \quad C \equiv p \delta q - K = \text{const}, \quad p \equiv \frac{\partial L}{\partial \dot{q}}.$$

That is the whole engine. The conserved quantity C is the Noether charge: the generalized momentum p , which the Lagrangian supplies, times the direction δq in which the symmetry pushes the system.

Run it on the simplest possible case: a free particle of mass m in one dimension, $L = \frac{1}{2}m\dot{x}^2$. Shift its position by a constant, $x \rightarrow x + s$. Velocities are untouched, so L does not change — an exact symmetry for every s , because there is no x in L to begin with. Here $\delta x = 1$ and $K = 0$, and the generalized momentum is $p = \partial L / \partial \dot{x} = m\dot{x}$, ordinary momentum. The theorem therefore returns $C = m\dot{x}$: momentum is conserved. There is no force to change it precisely because there is no x -dependence for a force to come from.

Then break the symmetry on purpose. Drop the same particle into a uniform gravitational field, $L = \frac{1}{2}m\dot{x}^2 - mgx$. Shift $x \rightarrow x + s$ again and this time L changes by $-mgs$, which is not the time derivative of anything, so the symmetry is gone. The theorem no longer promises conservation, and conservation indeed fails: Euler–Lagrange gives $\dot{p} = \partial L / \partial x = -mg$, which is Newton’s second law wearing a costume. Break the translation symmetry and you lose momentum conservation and acquire a force. They are the same event described twice.

The time-translation case is worth doing too, because it is the one Hilbert cared about. When L has no explicit t -dependence, the symmetry is $t \rightarrow t + s$, and the same manipulation yields a Noether charge with an extra term:

$$E = \dot{q} \frac{\partial L}{\partial \dot{q}} - L.$$

For the free particle this is $E = \dot{x}(m\dot{x}) - \frac{1}{2}m\dot{x}^2 = \frac{1}{2}m\dot{x}^2$ — the kinetic energy, as it must be. For the particle in gravity, L still has no explicit t , so $E = \frac{1}{2}m\dot{x}^2 + mgx$ is conserved even though momentum is not. One symmetry broken, another intact, and the theorem keeps exact track of which conservation law survives.

That bookkeeping is the practical content of the result, and it is the shape of the puzzle Hilbert handed Noether: to learn whether a quantity is conserved, stop hunting for the conservation law and go look for the symmetry. If the symmetry is broken, or does not exist in the form you assumed, the law breaks with it — not because physics failed, but because you were never entitled to it.

Global Symmetry, Local Symmetry, and the Second Theorem

Noether's 1918 paper has a less famous second theorem that turned out to be precisely what Hilbert needed, and it draws a sharp line between two kinds of symmetry. The symmetries in the table above are global: every point in space shifted by the same amount, every clock advanced by the same interval, everywhere at once. General relativity's core symmetry — general covariance, the freedom to relabel spacetime coordinates however you like — is different in kind. It is a local symmetry, whose parameter can be chosen independently at every point in spacetime, an infinite-dimensional family rather than a one-parameter one.

Noether's second theorem says that this richer symmetry does not produce a simple conserved quantity the way the first theorem does. Instead it produces a family of identities that the field equations must automatically satisfy [4], [6]. In general relativity that identity is the contracted Bianchi identity, a piece of pure differential geometry that falls out of the structure of curved spacetime and quietly guarantees the internal consistency of Einstein's equations [4].

That was the resolution, and it was not the one anyone expected going in. General relativity was not broken and energy had not vanished. What had broken was the older assumption that “energy conservation” must always mean one simple global bookkeeping law of the kind an exact global time-translation symmetry provides. Gravitational energy in general relativity is real, but it is not a tidy, sharply localized quantity that can always be added up cleanly across curved spacetime the way it can in flat, static spacetime — a subtlety that still generates careful technical argument in contexts such as the energy carried away by gravitational waves [4].

The Same Logic on a Bigger Stage

The most direct living descendant of the first theorem is the Standard Model of particle physics. Every conservation law particle physicists rely on daily — electric charge, the color charge of the strong force, weak isospin — comes from the recipe in the table above, applied to quantum fields rather than a single particle's coordinate: each force is built by demanding invariance under a local version of a symmetry group, and Noether's machinery converts that requirement into the corresponding conserved charge [2], [8]. Gauge symmetry, the organizing principle of essentially all particle physics since the 1970s, is Noether's theorem run in reverse. Instead of finding a symmetry and deriving a conservation law, physicists specify the conservation law they want and go looking for the symmetry that would produce it.

And the same logic that resolved Hilbert's 1915 puzzle turns up at the largest scale there is.

WORK OUT WHERE THE ENERGY COMES FROM

Two distant galaxies drift apart on the expanding cosmic background, gaining gravitational potential energy relative to each other, with nothing doing work on either of them. Which step of the derivation above tells you whether that ledger has to balance?

The step is the hypothesis. Our universe, described by the Friedmann–Lemaître–Robertson–Walker metric of modern cosmology, is not static — space itself is expanding, so the universe genuinely looks different from one cosmic moment to the next [2]. There is no exact time-translation symmetry to invoke, which means the theorem never promised a globally conserved energy for the universe in the first place, and none is observed. The two galaxies gain potential energy with no compensating input from anywhere, which looks like a violation of energy conservation exactly because, in the strict global sense, it is one [2]. This is not a fringe claim. It is the textbook-consistent consequence of applying the very theorem Hilbert commissioned to a universe that, on the largest scales, has no moment resembling any other.

What Was Actually Lost

From 1919 to 1933 Noether rebuilt a second field almost as a second career. Her work on ideal theory and abstract rings became so foundational that “Noetherian ring” is standard vocabulary in every graduate algebra course, taught to students who often have no idea it belongs to the same person as the physics [3]. In April 1933 the Nazi government dismissed her from Göttingen for being Jewish, with no pension and no severance. She wrote to a colleague that it was “much less terrible for me than for many others” [3] — a line that reads less like equanimity than like someone who had spent eighteen years being systematically underpaid and under-titled and had, by necessity, recalibrated what loss felt like. She sailed for Bryn Mawr College in Pennsylvania that October, lectured there and at Princeton’s Institute for Advanced Study through the following winter, and died suddenly in April 1935, at 53, from complications following what should have been routine surgery [3].

Year	Event
1907	Doctorate, Erlangen, under Paul Gordan
1915	Invited to Göttingen; faculty blocks the post
1918	Both theorems published
1919	Habilitation finally granted, age 37
1933	Dismissed by the Nazi regime; emigrates to Bryn Mawr
1935	Dies, age 53; Einstein’s tribute in the New York Times
1970s	Gauge symmetry becomes the Standard Model’s core

Einstein wrote to the New York Times days after her death that “in the judgment of the most competent living mathematicians, Fräulein Noether was the most significant creative mathematical genius thus far produced since the higher education of women began” [9] — a sentence that is, depending on how charitably it is read, either the highest compliment of the era or a faint one qualified almost to the point of vanishing. Both readings have been argued in the historical literature, and the ambiguity says more about 1935 than about her [9].

What makes this more than a biography with a happy mathematical ending is how completely the theorem detached itself from the circumstances of its birth. Nobody doing particle physics today needs to know that it came out of a specific argument between Hilbert and a skeptical universe about whether general relativity was self-consistent. The theorem simply is the connective tissue between the deepest structural fact about a physical theory — its symmetries — and the quantities experimentalists can measure and never see change. It converted three hard-won nineteenth-century discoveries into one twentieth-century insight, then kept paying dividends for a hundred years, from the Standard Model's gauge groups to the reason a cosmologist will say, without hesitation, that asking where the energy goes as the universe expands is a bit like asking what lies north of the North Pole: a question built on an assumption, time-translation symmetry, that the setting does not have.

Which leaves the harder question the theorem raises rather than settles. If every conservation law is a symmetry in disguise, then the symmetries we happen to have found are a map of what physics currently knows how to look for — and the deepest one, the invariance of the laws under the passage of time, turns out to fail for the only universe we actually inhabit. It is not obvious what else on the list is merely a very good local approximation.

REFERENCES

- [1] N. Wolchover, "How Noether's theorem revolutionized physics," *Quanta Magazine*, Feb. 7, 2025.
- [2] E. Siegel, "True: the expanding universe doesn't conserve energy," *Big Think — Starts With A Bang*.
- [3] J. J. O'Connor and E. F. Robertson, "Emmy Amalie Noether," *MacTutor History of Mathematics Archive*, University of St Andrews.
- [4] K. A. Brading and H. R. Brown, "Noether's theorems and gauge symmetries," and related historical treatments of the 1915–1918 Hilbert–Einstein energy dispute in general relativity; see also E. Noether, "Invariante Variationsprobleme," *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, 1918, pp. 235–257.
- [5] "We are a university, not a bath house," *FEBS Network*, on the Hilbert remark to the Göttingen faculty senate concerning Noether's habilitation (wording as commonly quoted; precise original sourcing is thin, as with many oft-repeated scientific one-liners).
- [6] J. Baez, "Noether's theorem in a nutshell," University of California, Riverside.
- [7] H. Goldstein, *Classical Mechanics*, and standard treatments of Lagrangian mechanics and the principle of least action, consistent with the derivation given here.
- [8] "Emmy Noether: the mathematician who changed the face of physics," *CERN EP News*.
- [9] A. Einstein, letter to the editor, *The New York Times*, May 4, 1935, reprinted in the *MacTutor History of Mathematics Archive*; on the debated reading of Einstein's phrasing, see "Can we please stop (mis)quoting Albert on Emmy, it's demeaning?," *The Renaissance Mathematicus*, 2017.

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